



Synergetic promotional effect of oxygen vacancy-rich ultrathin TiO₂ and photochemical induced highly dispersed Pt for photoreduction of CO₂ with H₂O

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ABSTRACT

The ultrathin TiO₂ nanosheets (TiO₂-U) with abundant defects were successfully fabricated as substrates to support well-dispersed Pt nanoparticles with low metal loading by photochemical route (P), with impregnated Pt/TiO₂-U, Pt/TiO₂-bulk and photoreduced Pt/TiO₂-bulk as control samples, to focus on studying the inductive effect of oxygen vacancy-rich ultrathin TiO₂ on highly dispersed Pt nanoparticles, and their synergetic promotional effect for CO₂ photoreduction. As expected, P-Pt/TiO₂-U exhibited excellent photocatalytic efficiency for the selective conversion of CO₂ into CH₄ and CO. The ultrathin TiO₂ nanosheets with a large number of low coordinative sites and ultralarge surface area promoted the rate of electron-transfer. The ultrafine Pt nanoparticles induced by photochemical strategy facilitated the efficient separation of electron-hole pairs. Furthermore, the synergy of metal and support improved the adsorption ability of CO₂. These three factors were considered to drive jointly the enhancement of catalytic performance in this system. This work offers deep insights for the design of highly efficient catalysts with coordinatively unsaturated sites for CO₂ photoconversion in the presence of H₂O.

1. Introduction

As the combustion of fossil fuels increasing, anthropogenic carbon emissions continues to increase, which is historically considered as the primary drive of global warming. In an attempt to meet this challenge, various strategies are currently proposed including the capture, storage [1] and catalytic hydrogenation of CO₂ [2–6]. The latter is a preferred approach for large-scale CO₂ utilization since this can yield high-value chemicals and fuels, making the process economically feasible and environmentally benign, though technically very demanding due to the high stability of CO₂ [7]. Developing highly efficient materials for photocatalytic CO₂ reduction is reported to be favorable to solve this problem, typically based on semiconductor (TiO₂ [8], ZnO, CdS, nanocomposites [9–11] and organic compounds) or supported metal-based catalysts [12]. However, the catalytic performance for CO₂ photoreduction with H₂O is still low, limited by the rapid recombination rate of electron-hole in the materials [13] and high barrier for activating the stable CO₂ molecules on the surface of catalysts [14]. Additionally, the reduction of H₂O to generate H₂ also proceed due to the lower reduction potential (0 eV) relative to the standard reduction potential of CO₂ to generate CO₂^{•−} (E = −1.9 eV vs the standard

hydrogen electrode) [15,16], and therefore compete with the reduction of CO₂ to CH₄ or CO.

Defects play an irreplaceable role in a variety of fields, including heterogeneous catalysis [17,18], photodegradation of antibiotics [19,20], gas sensors [21,22] due to its unique properties and remarkable performance. Some proposals discovered that the defective sites on metal oxide such as In₂O₃, TiO₂, ZnO and Ga₂O₃ based catalysts, promote the efficiency of the conversion of CO₂ considering the enhanced adsorption/activation ability of CO₂ and the influence on the separation of electron-hole [23–25]. For example, Liu and coworkers treated Cu/TiO₂ catalyst in H₂ to engineer more oxygen vacancies, and discovered that this catalyst exhibited improved efficiency of photoreduction CO₂, which was ten times higher relative to that of Cu/TiO₂ before treatment [26]. Moreover, a series of TiO₂ nanoparticles with single-electron trapped vacancy in the bulk and/or surface vacancy fabricated by Yang group [27]. They found that the coexistence of bulk and surface vacancies improved the photoreduction efficiency since both of them could enhance the ability of light absorption, while the latter helped the electrons-holes separation. In addition to atmosphere processing, the regulation of photomaterial thickness was also proceed as an effective mean to controllably create oxygen vacancies. Zhao et al. [28]

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successfully fabricated ultrathin ZnAl-LDH nanosheets possessing abundant low coordinative sites. Compared with the bulk, this ultrathin sample displayed outstanding activity for photoreduction of CO_2 , which was due to that coordinatively unsaturated Zn^{2+} - V_o promoted adsorption of reactants and inhibited the recombination of photoinduced charge. Additionally, the ultralarge surface area ensured the access of reagents to active sites without limitation of diffusion. Considering that the ultrathin materials possess not only rich defects, but also large surface area, this inspires us to thin the universal TiO_2 materials, which will make it more widely applied.

Except for semiconductors, the introduction of active metal, especially surface platinization is believed to serve not only as sinks to trap the electrons, thus retard the recombination of electron-hole [29–31], but also as the active sites for activation of CO_2 molecules [32]. However in previous studies, Pt nanoparticles were historically fabricated by wet chemical method, in which the size was difficult to control [33], and consequently led to the formation of large Pt nanoparticles considered as recombination center [34]. As a result, reducing the particles size of active metal is also crucial for the improvement of photoreduction efficiency. Recently, a photochemical strategy at room-temperature to fabricate a highly stable, atomically dispersed metal catalyst developed by Zheng et al. [35] draws wide attention, which provided convenient, controllable and environmentally benign route to obtain highly dispersed catalysts.

Herein, in this paper, by intelligently combining the advantages of ultrathin TiO_2 and photochemical strategy, we first fabricated the TiO_2 with thicknesses approaching atomic scales supported the ultrafine Pt nanoparticles with low metal loading by photochemical strategy, and focus on studying the influence of the preparation method and support thicknesses on the photocatalytic behaviors of Pt/ TiO_2 . As expected, the highly dispersed Pt nanoparticles on the ultrathin TiO_2 demonstrated superior photocatalytic efficiency for the selective conversion of CO_2 into CH_4 and CO with $666.6 \mu\text{mol g}^{-1} \text{h}^{-1}$ of total electronic yield. More importantly, after three cycling tests, this sample also exhibited superior operational stability. This enhanced performance was attributed to the following factors: (1) the ultrathin TiO_2 nanosheets with abundant defects and ultralarge surface area promoted the rate of electron-transfer; (2) the ultrafine Pt nanoparticles facilitated the separation of photogenerated electrons-holes, and thus improved the light-harvesting capacity and quantum efficiency, confirmed by ESR, XPS, PAS, UV–vis and PL analysis; (3) the synergy of metal and support improved the adsorption ability of CO_2 . This work attempts to provide insights for the design of highly efficient catalysts of CO_2 conversion in the presence of H_2O .

2. Experimental

2.1. Materials

Hydrochloroplatinic acid ($\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$), sodium hydroxide (NaOH), hydrochloric acid, ethanol and commercial titanium dioxide (TiO_2 -B) were purchased from Aladdin. Titanium tetrachloride (TiCl_4), ethylene glycol (EG) were bought from Alfa Aesar. The water used in all experiments was deionized. All chemicals were used without further purification.

2.2. Preparation of ultrathin TiO_2

The preparation of ultrathin TiO_2 was similar with the previous report by Wang et al. [36]. In detail, 0.05 mmol TiCl_4 and 30 mL EG were mixed and stirred until HCl disappears. Then, 1 mL water was added to the mixed solution, followed by aging at 150°C for 24 h in the stainless steel autoclave of 100 mL. The obtained white products were washed by ethanol, and dried at room temperature in vacuum oven. The obtained samples were denoted as TiO_2 -U.

2.3. Preparation of Pt/ TiO_2

50 mg of the dried ultrathin TiO_2 or commercial TiO_2 (TiO_2 -B) was dispersed in 10 mL of aqueous solution containing 75 μL H_2PtCl_4 solution (18 mmol/L) under stirring for 30 min, and further treated with UV light for 10 min [35]. The grey product was washed by water and dried in vacuum oven. The samples were denoted as P-Pt/ TiO_2 -U and P-Pt/ TiO_2 -B, respectively. For comparison, TiO_2 -U and TiO_2 -B supported Pt catalysts with the same Pt content were prepared by the impregnation method followed by successive reduction at 450°C and denoted as I-Pt/ TiO_2 -U and I-Pt/ TiO_2 -B.

2.4. Characterization of catalysts

The patterns of X-ray diffraction (XRD) were collected using a Shimadzu XRD-600 X-ray powder diffractometer with $\text{Cu K}\alpha$ ($\lambda = 0.154 \text{ nm}$) as radiation source. Sample morphologies and particle sizes were measured using a JEOL-2100 F high-resolution transmission electron microscope (HRTEM). The thickness of samples were examined using a Digital Instruments Version 6.12 atomic force microscope (AFM). Brunauer-Emmett-Teller (BET) method was employed to calculate the surface area of the samples using a Micromeritics Gemini VII2990 instrument with N_2 as the adsorbate at 77 K. The loading of active metal was detected by a Shimadzu ICPS-7500 inductively coupled plasma emission spectrometer (ICP-AES). The ultraviolet-visible diffuse reflection absorption spectra (UV–vis) were collected by an UV–vis spectrometer (PGENER-AL TU-1901, Beijing) with BaSO_4 as a reference. To investigate the concentration/type of oxygen vacancies, positron annihilation spectroscopy (PAS) using a fast-slow coincidence ORTEC system (a time resolution of 187 ps for the full width at half maximum) and Raman measurements excited at 532 nm laser were conducted. The photo-luminescence (PL) spectra were identified on a fluorescence spectrometer (Shimadzu; UV-3600). The electronic structure of the compositions on the surface of catalysts was tested by a Thermo VG ESCALAB 250 spectrometer (Al $\text{K}\alpha$ anode) with C1s (284.6 eV) as calibration peak. Electron spin resonance spectra (ESR) were collected on a JEOL JES-FA200 spectrometer at 90 K. In situ Fourier Transform Infrared (FTIR) spectroscopy of CO and CO_2 were recorded on a Bruker Tensor 27 instrument. Prior to experiments, the samples were pre-treated in He at 100°C for 60 min, followed by recording a background with a resolution of 4 cm^{-1} . For in situ FTIR of CO , the catalysts were exposed to CO flow for 30 min and collected the spectra under the different pressure. For in situ FTIR of CO_2 , CO_2 was first purged under dark. Then, the catalysts were irradiated under UV–vis light (380–800 nm) for 30 min and recorded the spectra. In the term of FT-IR for CO_2 reacted with H_2O , CO_2 flow with H_2O vapor were inlet under dark, followed by recording a background with a resolution of 4 cm^{-1} . Then, the sample was irradiated with UV–vis light, and the spectrum was recorded.

2.5. Photocatalytic reduction of CO_2

The reaction was conducted in a custom-made gas phase reactor (50 cm^3 of chamber volume; quartz window on the top of chamber) in Fig. S1. 10 mg of samples were spread in the chamber with 0.08 MPa CO_2 (Beijing Beiweng Gas Comp. 99.999%). Water vapor was added to the chamber through CO_2 bubbling. Light was irradiated using a 300 W Xe lamp as the light source. The isotope-labeled experiments were carried out using $^{13}\text{CO}_2$, the amount of products (CH_4 , CO and H_2) was analyzed using gas chromatograph (Shimadzu; GC-2014C) equipped with thermal conductivity detector (TCD) and flame ionization detector (FID).

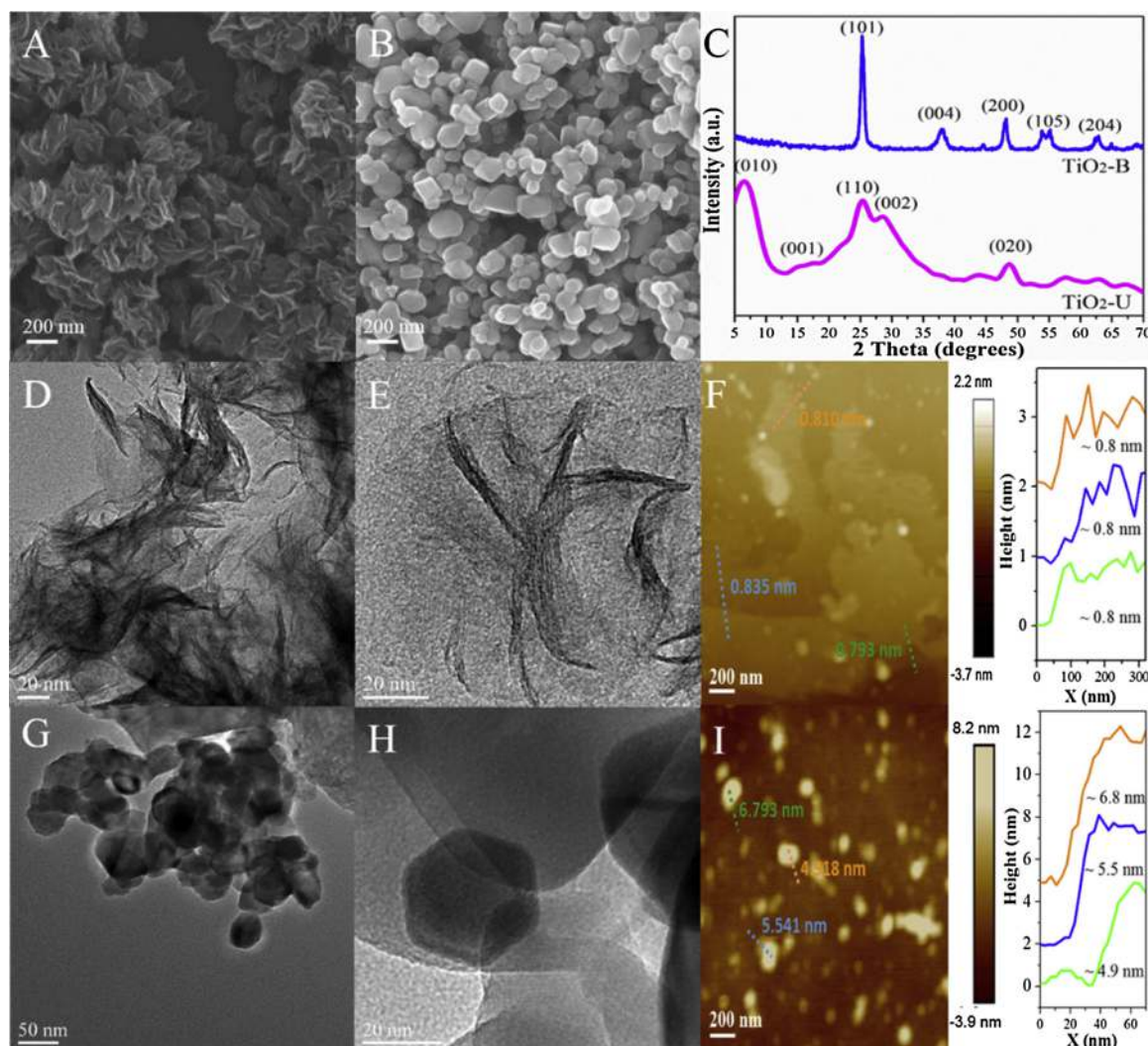


Fig. 1. SEM image of (A) ultrathin TiO₂ nanosheets and (B) bulk TiO₂; (C) XRD patterns; HRTEM image of (D, E) ultrathin TiO₂ nanosheets and (G, H) bulk TiO₂; AFM images of (F) ultrathin TiO₂ nanosheets and (I) bulk TiO₂.

3. Results and discussions

3.1. Structure and morphology of ultrathin TiO₂

Ultrathin TiO₂ nanosheets were fabricated with the commercial TiO₂ as reference. As shown in XRD patterns (Fig. 1), the main reflections detected in commercial TiO₂ are indexed as (101), (004), (200), (105) and (204) planes, showing a typical anatase TiO₂ phase in line with JCPDS 21-1272. Different from the anatase, the peaks of the TiO₂ nanosheets correspond to (110), (002) and (020), which are consistent with the ultrathin materials in reference [36] (space group C2/m, JCPDS 74-1940). It is worth noting that much intense diffraction peak of (020) is observed relative to that of (110), suggesting that this facet is preferentially exposed [35]. Additionally, no other peaks are observed, indicating that this material is highly pure. The lattice parameters are further calculated using Rietveld refinement, and the values of lattice parameters are listed in Table 1. To better understand the morphology of ultrathin nanosheets, electron microscopy (SEM and HRTEM) was employed. From the images in Fig. 1 and Fig. S2, the structure of ultrathin TiO₂ is observed to be in sheet-like and the lateral size is 30–50 nm, while the commercial samples exhibit hexagonal particle shape with a larger particle size ($\approx$ 50–200 nm). Considering that the thickness of samples can not be accurately measured by electron microscopy, we further used AFM analysis to obtain the relative

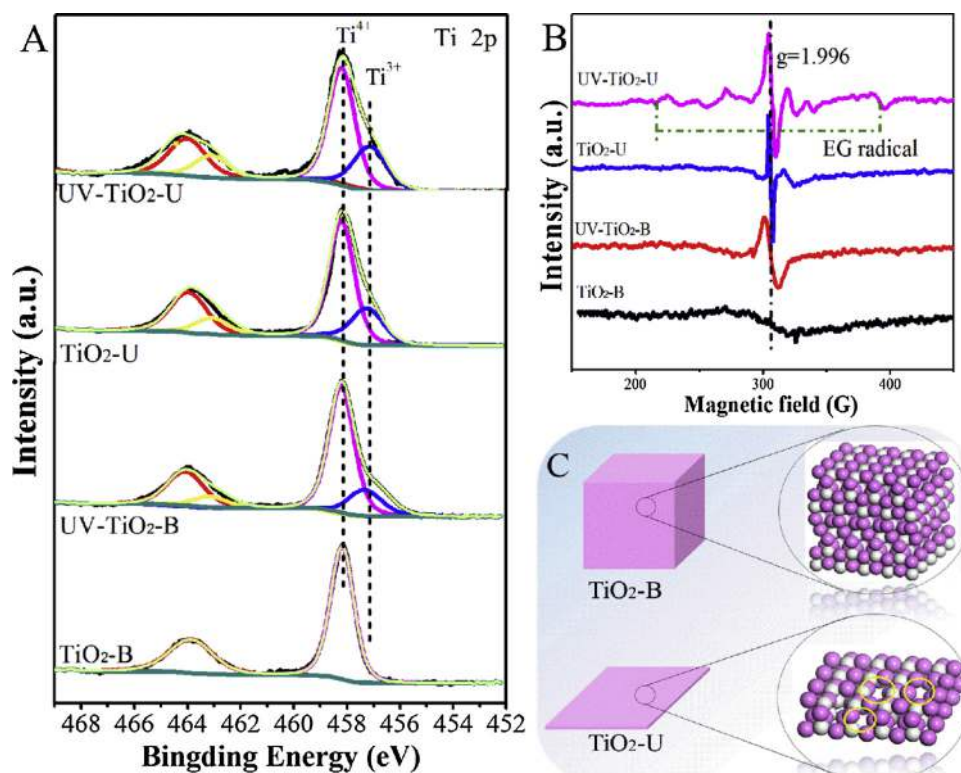
information in several different zones. For the TiO₂ nanosheets, the platelet thickness is ca. 0.8 nm roughly equal to the thickness of two atomic layers, while the bulk TiO₂ shows a larger value of 6.8 nm in thickness. Combined with the above characterization, this data affirm the successful synthesis of two-atom-thick ultrathin TiO₂. The curves of N₂ adsorption/desorption in Fig. S2 for both of materials are consistent with the type IV isotherm, depicting the adsorption of micropores and mesopores with pore size of 3–20 nm. Surprisingly, the specific surface area of ultrathin TiO₂ is estimated to be 473 m² g^{−1}, 26-fold larger than that of bulk TiO₂ (18 m² g^{−1}). The higher surface area is expected to favor the dispersion of active metal and the adsorption of reactant molecules, benefiting the enhanced photocatalytic activity [37].

3.2. Qualitative and quantitative characterization of surface defects

It is well documented that ultrathin nanosheets possess different properties such as atom arrangements and surface defects [38,39], which are investigated using XPS analysis. The peaks at 458.2 and 464.0 eV are ascribed to Ti⁴⁺, whereas the peaks at 457.2 and 463.0 eV are assigned to Ti³⁺. As shown in the Ti 2p XPS spectra (Fig. 2A), the main valence of Ti element in the bulk is +4, while the ultrathin TiO₂ exhibits higher content of Ti³⁺, the formation of which is in proportion to the oxygen vacancies. This indicates a larger number of defects in the ultrathin TiO₂. After introducing UV light for 10 min, the ratio of Ti³⁺ /

Table 1The parameters of lattice, surface area and pore structure of TiO₂ supports.

Samples	Lattice parameter (Å/°)	Surface area (m ² ·g ⁻¹)	Pore volume (cm ³ /g)	Pore size (nm)	Ti ³⁺ /Ti ⁴⁺ (%)
TiO ₂ -B	a = 3.784; b = 3.784 c = 9.515; α = β = γ = 90°	18	0.09	18.0	–
TiO ₂ -U	a = 12.174; b = 3.743 c = 6.526; β = 107.3°	473	0.43	3.7	26.3
TiO ₂ -B-UV	–	–	–	–	14.9
TiO ₂ -U-UV	–	–	–	–	35.9

**Fig. 2.** (A) XPS spectra (B) ESR spectra of the ultrathin TiO₂ nanosheets and bulk TiO₂ before and after UV irradiation; (C) Scheme of coordinatively unsaturated ultrathin TiO₂ nanosheets.**Table 2**

Positron lifetime and relative intensities of samples.

Samples	τ ₁ (ns)	τ ₂ (ns)	τ ₃ (ns)	I ₁ (%)	I ₂ (%)	I ₃ (%)	I ₂ /I ₁
TiO ₂ -B	0.1829	0.3629	2.538	36.0	59.6	1.438	1.65
TiO ₂ -U	0.1894	0.3931	2.518	35.6	62.9	1.601	1.77
P-Pt/TiO ₂ -B	0.1947	0.3897	2.652	43.0	55.3	1.647	1.28
P-Pt/TiO ₂ -U	0.1818	0.4362	2.519	31.8	66.6	1.641	2.10
I-Pt/TiO ₂ -B	0.1887	0.3657	2.393	36.6	61.8	1.654	1.68
I-Pt/TiO ₂ -U	0.1878	0.3997	2.385	33.5	64.9	1.567	1.94

Ti⁴⁺ (Table 1) obviously rise for both ultrathin and bulk TiO₂, suggesting that more electrons are generated under light and trapped in Ti 3d orbitals to form Ti³⁺ sites [40], but the ratio of Ti³⁺/Ti⁴⁺ in former is always larger than latter.

ESR analysis is a powerful technique to detect and study the materials with unpaired electrons. The surface oxygen vacancies have two electrons (Vo^{••}) or no electron (Vo), which don't show any signals in ESR analysis. Under the certain conditions, the oxygen atoms were removed, leaving two electrons in per oxygen vacancy, some of which were captured by Ti-3d orbitals of the neighboring Ti atoms to form Ti³⁺ species [41]. As a result, the existence of Ti³⁺ can verify the formation of surface oxygen vacancies indirectly. Fig. 2B shows the spectra of ESR

over the four materials. It can be seen that the dominated peak at $g = 1.996$ in all the samples matches perfectly with the Ti³⁺ specie [42]. In accordance with above results, the intensity of the peak in ultrathin TiO₂ is much higher than that of bulk TiO₂, suggesting the ultrathin material possesses the abundant surface vacancies. After UV–vis irradiation ($\lambda < 400$ nm), the signal intensity of both TiO₂ materials significantly increase, indicating the improved concentration of vacancies. The other small peaks in the ultrathin TiO₂ are assigned to EG radicals (HOCH₂CHOH), appearing after UV–vis irradiation, which originates from that the holes generated on TiO₂ nanosheets broke the bonds between glycolate and TiO₂ [43]. This observation was also found by Wang et al. [36] who fabricated the ultrathin TiO₂ using EG, further confirmed the stability of EG radicals by TG measurement. A similar stability is also confirmed to exist in our system (Fig. S3).

To further quantitatively achieve the size, type (τ_1 , τ_2 and τ_3) and relative density (I_1 , I_2 and I_3) of various unsaturated sites, namely defects, PAS as a powerful tool to study defects in solids was carried out, and the values for the pure TiO₂ are listed in Table 2. The shortest-lived one (τ_1) arises from smaller monovacancies in the bulk section (such as oxygen vacancy) [44,45]. The intermediate components is ascribed to positrons captured by the larger vacancy-type defects such as the Ti³⁺–oxygen vacancy associates on the surface. Such defect type can serve as the center for trapping positron, and the low electron density

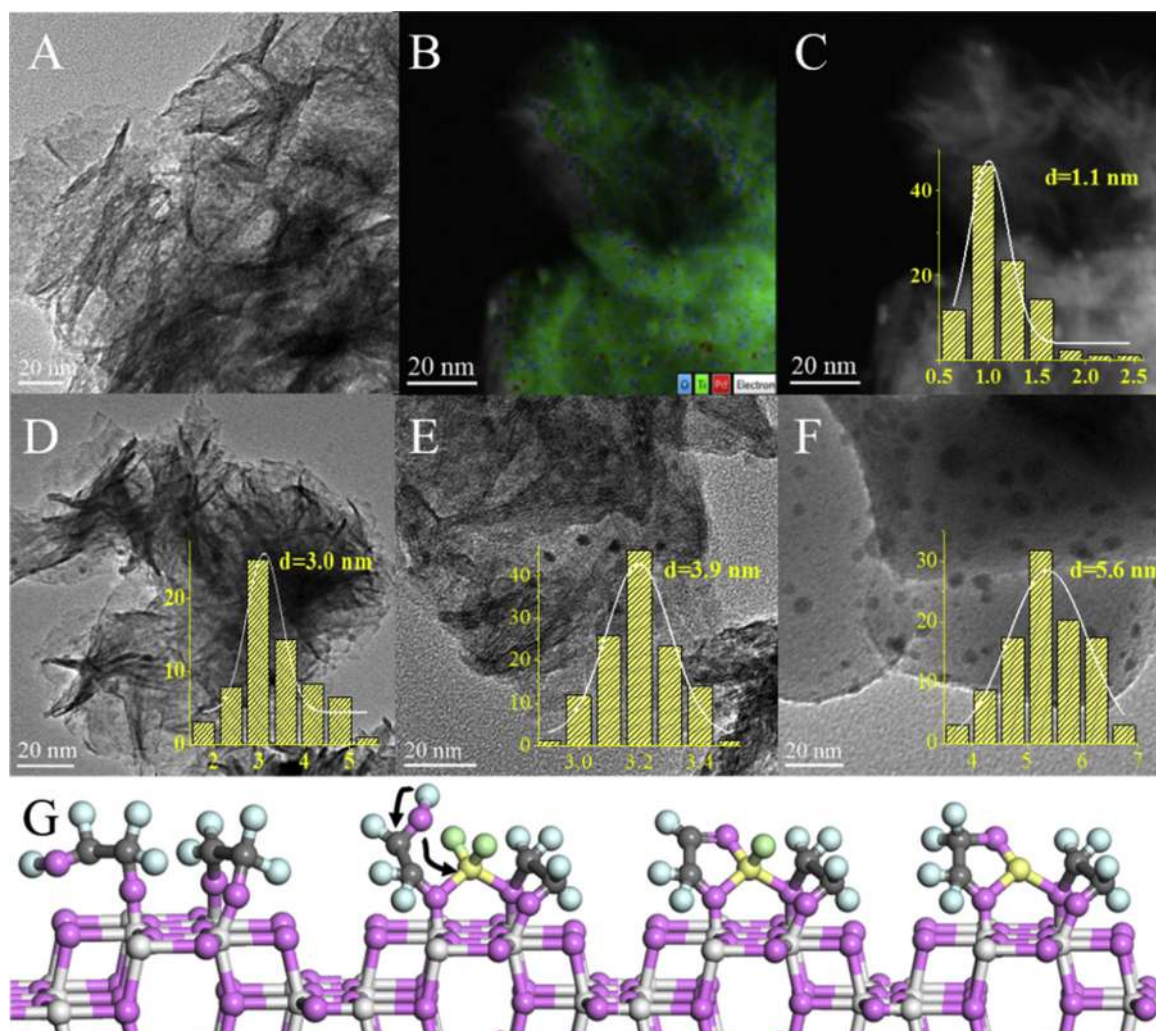


Fig. 3. (A–C) HRTEM images and STEM-HAADF-mapping of P-Pt/TiO₂-U; HRTEM images of (D) I-Pt/TiO₂-U, (E) P-Pt/TiO₂-B, (F) I-Pt/TiO₂-B. The insets show histograms of the particle size frequency distribution. (G) The mechanism of the dispersion of Pt nanoparticles induced by EG radicals.

reduces the positrons annihilation rate, leading to increase of the life-time relative to the τ_1 [46,47]. The longest component (τ_3) is attributed to the annihilation of ortho-positronium atoms in the large voids in nanoscale [48,49].

Well-know that surface defects not only serve as the sites for adsorbing charge, but also as the traps for capturing carrier, which migrate to the adsorbed species, and consequently limit the electrons and holes recombination [50], while defects in the bulk only act as capture centers for the photogenerated carriers where electrons and holes recombine [51]. Consequently, increasing the surface defects is helpful to enhance the efficiency for photocatalysis. The relative intensities of the positron lifetimes determine the information about the concentration of these defects. In our study, the ratio of I_2 (surface defects)/ I_1 (bulk defects) of TiO₂-U (1.77) is higher than that of TiO₂-B (1.65), reflecting the higher proportion of the surface Ti^{3+} –oxygen vacancy in the ultrathin TiO₂, which appears to be favor to the electron-transfer rate. A possible structure of coordinatively unsaturated ultrathin TiO₂ nanosheets is shown in Fig. 2C.

3.3. Characterization of ultrathin TiO₂ supported Pt catalysts

TiO₂ with thicknesses approaching atomic scales play a determinant role in the dispersion and stabilization of active metal nanoparticles as it possesses the abundant defects and large surface area. Once exposed to UV, electron-hole pairs were generated, in which electrons were

captured in Ti-3d orbitals to generate Ti^{3+} species, and holes contributed to the $-\text{OCH}_2\text{CHOH}$ formation, confirmed by XPS and ESR. Herein, we employed the ultrathin TiO₂ covered by deprotonated EG as support, and intelligently combined with photoreduction as preparation method to fabricate the ultrafine Pt nanoparticles. As shown in Fig. 3 and Fig. S4, the morphology for the ultrathin TiO₂ nanosheets is well preserved after the incorporation of active species. 200 particles in different regions (Fig. 3 insets) are randomly selected to measure the mean size and distribution. For photoreduced Pt/TiO₂-U, the average size is found to be 1.1 nm, which is smaller than those of photoreduced Pt/TiO₂-B (3.9 nm), impregnated Pt/TiO₂-U (3.0 nm) and Pt/TiO₂-B (5.6 nm). Additionally, the metal dispersion of all catalysts was surveyed by CO-chemisorption, although the adsorption of CO on Pt atoms is complicate and controversial. But the dispersion trend of all catalysts (Table S1) in the sequence of P-Pt/TiO₂-U > I-Pt/TiO₂-U > P-Pt/TiO₂-B > I-Pt/TiO₂-B is consistent with HRTEM analysis, demonstrating the Pt nanoparticles on the ultrathin TiO₂ is well-dispersed induced by photochemical strategy. Two drives should be responsible for the ultrafine particle size. In the process of preparation (Fig. 3G), UV light-induced EG radicals on TiO₂ nanosheets facilitates to remove the Cl^- on Pt precursor and form the Pt-O bonds, and thus highly stable dispersed Pt nanoparticles are achieved. Besides, the ultralarge surface area of TiO₂ is also a critical factor for preparing ultrafine Pt catalysts [52].

The IR spectra were determined with CO as adsorbed molecule to

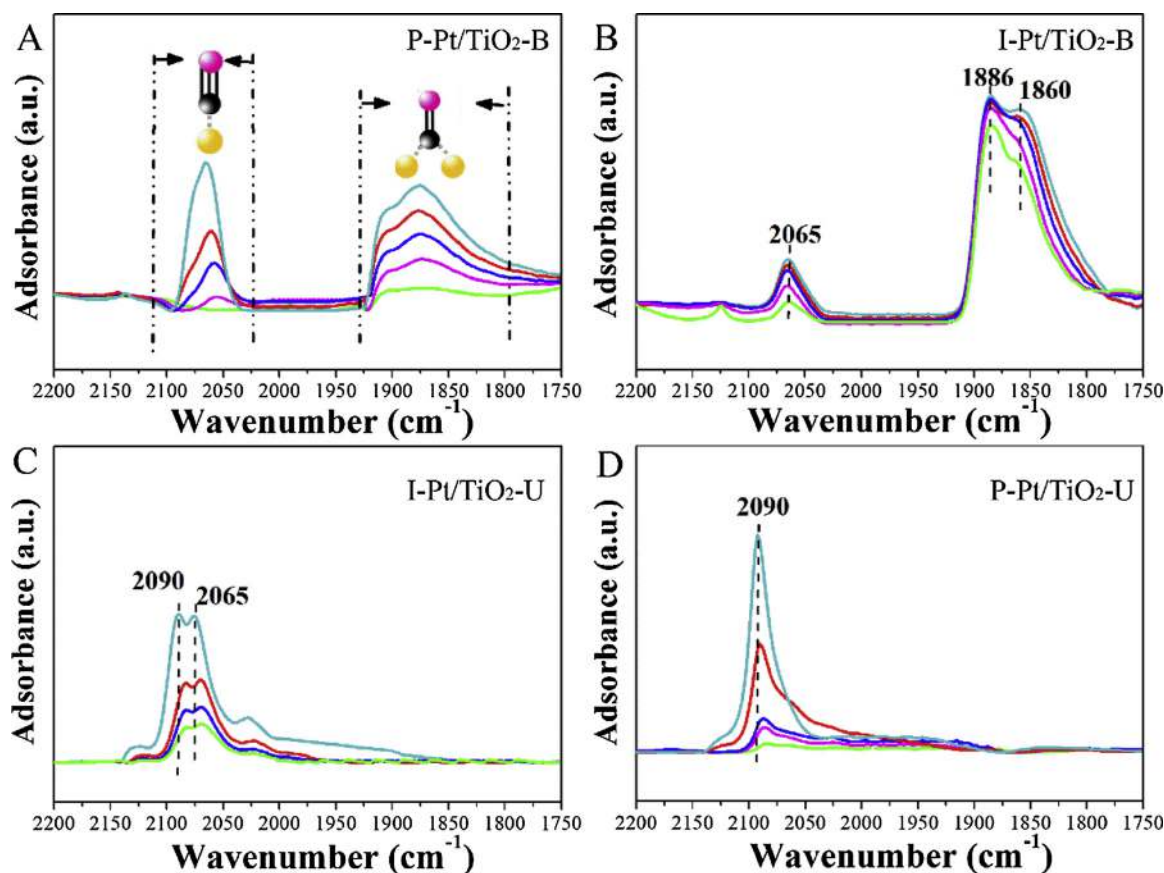


Fig. 4. CO-FTIR spectra of Pt catalysts under the decreasing pressure.

explore the surface structure of Pt atoms. Two bands in Fig. 4 are observed in P-Pt/TiO₂-B and I-Pt/TiO₂-B around 1860 cm⁻¹ and 2065 cm⁻¹, corresponding to the bridged bound CO and linearly coordinated CO on the Pt surface [53]. The decrease in the particle size of I-Pt/TiO₂-U and P-Pt/TiO₂-U lead to the increase of band intensity originated from the linear adsorbed CO at the expense of the bridged CO band, accompanied with the appearance of band at 2090 cm⁻¹ assigned to CO adsorbed on Pt terrace atoms. Summarizing the above results, it is concluded that the introduction of ultrathin TiO₂ and photochemical strategy facilitate to generate the largest number of Pt terrace atoms in the P-Pt/TiO₂-U catalyst. To investigate qualitatively and quantitatively the defects contained in Pt catalysts, PAS was also carried out over the catalysts, and the values are listed in the Table 2. The intensity ratio of I₂/I₁ for P-Pt/TiO₂-U catalyst is 2.10, higher than that of others, illustrating the predominance of large size defects (such as terrace-like defect and boundary-like defects) in the ultrathin TiO₂ supported Pt catalysts [27,45], in agreement with the IR data. It is well-known the low-coordinative sites are easier to activate the reactants, and act as trap centres for electrons, facilitating the photocatalytic CO₂ reduction.

3.4. The catalytic behavior of CO₂ photoreduction

The photocatalytic CO₂ reduction under UV light irradiation was employed to evaluate the catalytic behavior of ultrathin TiO₂ supported Pt catalysts derived by photochemical strategy, with I-Pt/TiO₂-U, P-Pt/TiO₂-B and I-Pt/TiO₂-B as control samples. The Pt contents of catalysts derived from ICP measurement (Table S1) are lower than theoretical value (0.5 wt.%) but in the error range. Fig. 5A and B illustrate the plot of CO and CH₄ moles over six samples versus reaction time. The amounts for both products linearly increase as the function of the time, confirming that this formation process of products proceeds in a steady-

state manner rather than a transient manner. This similar phenomenon is reported by Wang et al. [7]. At the reaction time of 10 h, pure bulk and ultrathin TiO₂ supports produce CO as the major product with amounts of 1.5 and 4.4 μmol, respectively. Upon introducing Pt nanoparticles on the bulk TiO₂, the product distributions of samples exhibit the differences. The amount of CH₄ improves from 0.3 μmol (TiO₂-B) to ca. 5.0 μmol (I-Pt/TiO₂-B and P-Pt/TiO₂-B). When the ultrathin TiO₂ is used to support active metal, the amount of CH₄ further dramatically increases to 47.0 μmol over P-Pt/TiO₂-U. The formation rate of CH₄ (Fig. 5D) is as follows: P-Pt/TiO₂-U (66.4 μmol g⁻¹ h⁻¹) > I-Pt/TiO₂-U (10.0 μmol g⁻¹ h⁻¹) > P-Pt/TiO₂-B (5.5 μmol g⁻¹ h⁻¹) > I-Pt/TiO₂-B (5.0 μmol g⁻¹ h⁻¹), in pace with the decrease of Pt particle size, which may contribute to the efficient electron-hole separation. Except for the formation of CH₄, CO with 54.2 μmol g⁻¹ h⁻¹ of the formation rate is also observed over the P-Pt/TiO₂-U. To track the carbon sources of products including CH₄ and CO, we further performed ¹³C isotopic label experiment. Fig. 5C displays the mass spectra of CH₄ and CO using ¹³CO₂ in place of ¹²CO₂ over P-Pt/TiO₂-U under same reaction conditions. The peaks at *m/z* = 29 and *m/z* = 17 associated with ¹³CO and ¹³CH₄ suggest that the origins of CH₄ and CO are indeed produced from the reactant CO₂ [54]. Moreover, we also observed the peaks at *m/z* = 16 and 15, which may be related to the fragment ions from ¹³CH₄ [55] or ¹²CH₄ (EG) and its corresponding fragment ions. To rule out the contribution of ethylene glycol on the formation CO or CH₄ in the reaction, the thermogravimetric analysis was performed over both fresh and used P-Pt/TiO₂-U catalysts (Fig. S3). As expected, the loss weight of used catalyst (15%) is basically consistent with that of fresh catalyst (14%), suggesting that EG on the surface of TiO₂-U is stable in the reaction, and thus confirm the detected CO or CH₄ formation through CO₂ reactant, instead of the ethylene glycol. According to the definition of the selectivity for CO₂ reduction in the previous references [7] (Selectivity for CO₂ reduction (%) = [2 r (CO) + 8 (CH₄)] / [2 r (CO) + 8 (CH₄) + 2

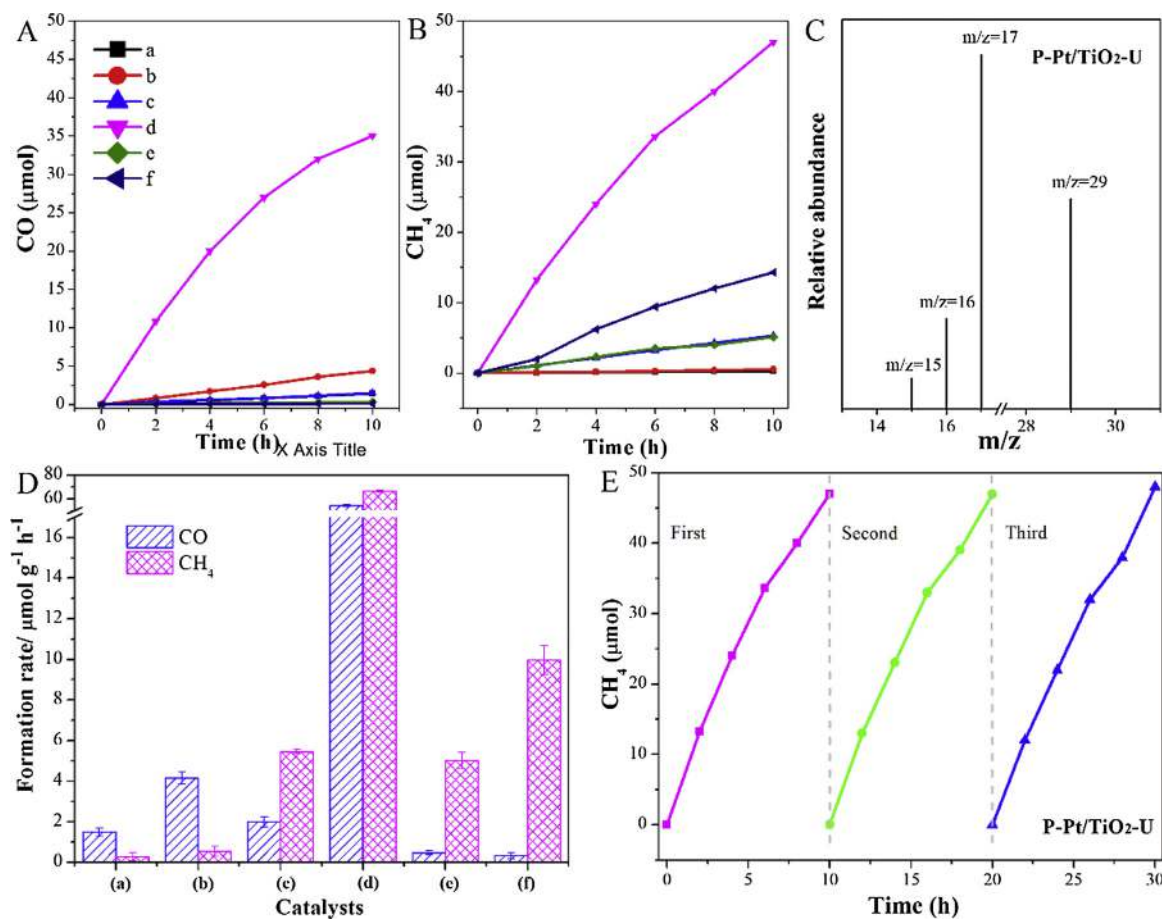


Fig. 5. Plots of (A) CO (B) CH₄ amount versus reaction time over (a) TiO₂-B, (b) TiO₂-U, (c) P-Pt/TiO₂-B, (d) P-Pt/TiO₂-U, (e) I-Pt/TiO₂-B and (f) I-Pt/TiO₂-U in the photocatalytic conversion of CO₂ in the presence of water vapor under UV light; (C) Mass spectra of ¹³CH₄ (*m/z* = 17, 16 and 15) and ¹³CO (*m/z* = 29) produced over P-Pt/TiO₂-U in photocatalytic reduction of ¹³CO₂; (D) photocatalytic CO and CH₄ evolution rates over (a) TiO₂-B, (b) TiO₂-U, (c) P-Pt/TiO₂-B, (d) P-Pt/TiO₂-U, (e) I-Pt/TiO₂-B and (f) I-Pt/TiO₂-U; (E) reusability of P-Pt/TiO₂-U.

Table 3

Catalytic performances of TiO₂ supports and the corresponding Pt catalysts in the photoreduction of CO₂ with H₂O.

Samples	TiO ₂ -B	TiO ₂ -U	I-Pt/ TiO ₂ -B	P-Pt/ TiO ₂ -B	I-Pt/ TiO ₂ -U	P-Pt/ TiO ₂ -U
<i>Y_e</i> (μmol g ⁻¹ h ⁻¹)	7.2	34.7	158.0	141.9	160.3	666.6
H ₂ (μmol g ⁻¹ h ⁻¹)	–	–	58.4	47.5	40.2	13.5
<i>τ_{average}</i> (ns)	27.4	29.2	32.0	39.8	120.2	143.9
CO ₂ uptake (μmol g ⁻¹)	5.7	27.1	16.3	20.8	39.6	52.2

(H₂) × 100%), 95.9% of selectivity is obtained for P-Pt/TiO₂-U, indicating only a small amount of by-product hydrogen exists. Such low hydrogen content suggests the H₂ evolution from H₂O reduction has been inhibited in a great extent, which is because that the most of active hydrogen are directly utilized to generate the CH₄ or CO owing to a high degree of separation of electrons and protons to avoid the H₂ production. To further analyze the efficiency of reduction, the total electronic yield described by *Y_e* is calculated involving all reductive products as following: *Y_e* = 2*r* (CO) + 8*r* (CH₄) + 2(H₂), where *r* is the formation rate of the corresponding products. 666.6 μmol g⁻¹ h⁻¹ of total electronic yield over P-Pt/TiO₂-U is more than 4 times higher than that of other catalysts (Table 3). The excellent efficiency of CO₂ reduction compares well with the activities of some widely used semiconductors for the photoreduction of CO₂ (Table S2). Furthermore, the reusability of P-Pt/TiO₂-U catalyst is performed. After 10 h time on stream, the reaction gases in the system was exhausted, and reactants

was inlet again (Fig. 5E). No prominent change of the activity is observed in the second and third runs.

3.5. Insight into the structure–property relationships

To give an insight into the nature of the excellent photocatalytic activity of ultrathin TiO₂ supported highly dispersed Pt nanoparticles, UV–vis diffuse reflectance spectra was exploited as a powerful technique to investigate the abilities of light absorption. As displayed in Fig. 6, the bare bulk TiO₂ exhibits the absorption edge at ~380 nm with band gap energy of 3.2 eV, implying that the only absorption is in UV region [56,57], while an effective shift of the absorption range to the visible light of 400 nm is noticed for the ultrathin TiO₂. After supporting active Pt metal, the absorptions of catalysts are estimated to be in the range of 400–520 nm, in which Pt/TiO₂-U prepared by photoreduction possesses the maximum value (ca. 513 nm) of absorption edge, indicating the large capability of photoexcited charges [58].

Furthermore, the electronic structure of Pt nanoparticles was analysed by XPS measurement. The collected Pt 4f XPS spectra shown in Fig. 7A contains two peaks at 70.7 eV (Pt 4f_{7/2}) and 74.1 eV (4f_{5/2}) attributed to the Pt⁰ [59]. For the I-Pt/TiO₂-B, the peaks of Pt⁰ slightly shift to lower binding energy (BE) by approximately 0.3 eV compared to the monometallic Pt catalyst in the literatures [60], suggesting the electron transfer from TiO₂ to Pt nanoparticle. When the ultrathin TiO₂ is used to support Pt atoms, a further shift towards lower energy is noticed due to more Ti³⁺ species detected in the ultrathin TiO₂ [61]. Additionally, the change in electronic environment is also related to the size effect [62]. In terms of size effect, the Pt⁰ 4f_{7/2} peak in the case of P-

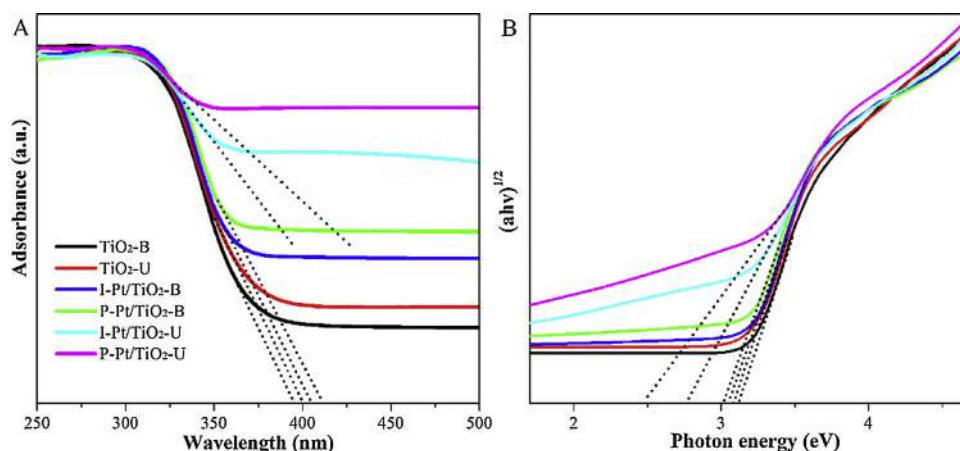


Fig. 6. (A) UV-vis DRS analysis; (B) Plot of $(ah\nu)^{1/2}$ versus photon energy (eV) of absorbed light for the calculation of band gap energy of all the samples.

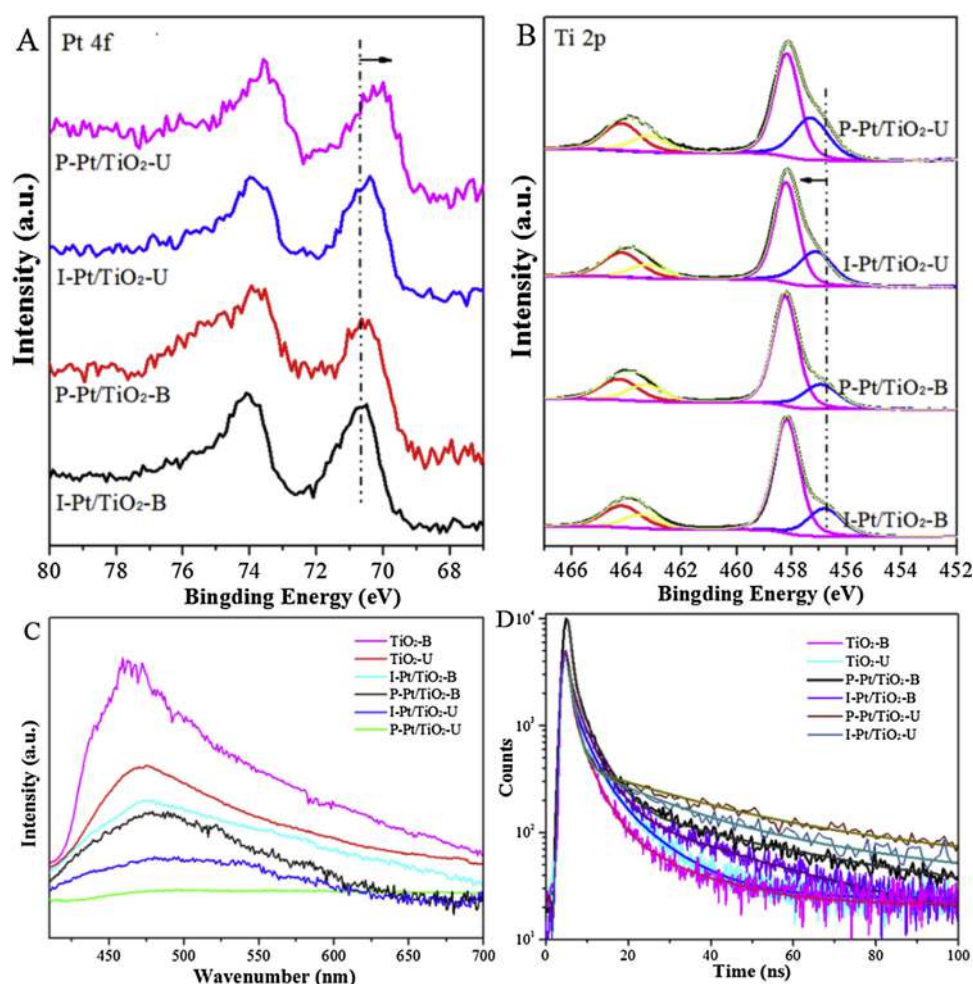


Fig. 7. (A) XPS Pt 4f spectra and (B) XPS Ti 2p spectra of Pt catalysts; (C) PL spectra under 365 nm excitation; (D) Fluorescence lifetime of all the photocatalysts.

Pt/TiO₂-U catalyst with smallest metal particle size should be positively shifted. However, it shifts to lower BE by about 0.4 eV compared to the P-Pt/TiO₂-B catalyst resulting from the stronger interaction between the TiO₂ and Pt nanoparticles which offsets the size effect. Along with the shift in Ti 2p XPS (Fig. 7B), we conclude that the electron indeed transfer from TiO₂ to the metals centres, which further affect the efficiency of photoreduction.

To confirm the claim, the ability of electron-hole recombination in the samples was also revealed by the photoluminescence (PL) emission

spectra. A band at ~ 470 nm for all the samples is observed, associated with the surface oxygen vacancies. Relative to the bulk TiO₂, the intensity of band over ultrathin TiO₂ slightly decrease (Fig. 7C). Upon Pt nanoparticles are deposited on the supports, the further reduction is obtained. These observations reveal a recombination rate of charge carrier [63], explained by the transfer of photogenerated electrons to the metals centres. It is worth noting that the Pt/TiO₂-U catalysts reduced by photochemical strategy with ultrafine Pt nanoparticles are more efficient to separate the electrons from holes compared with other

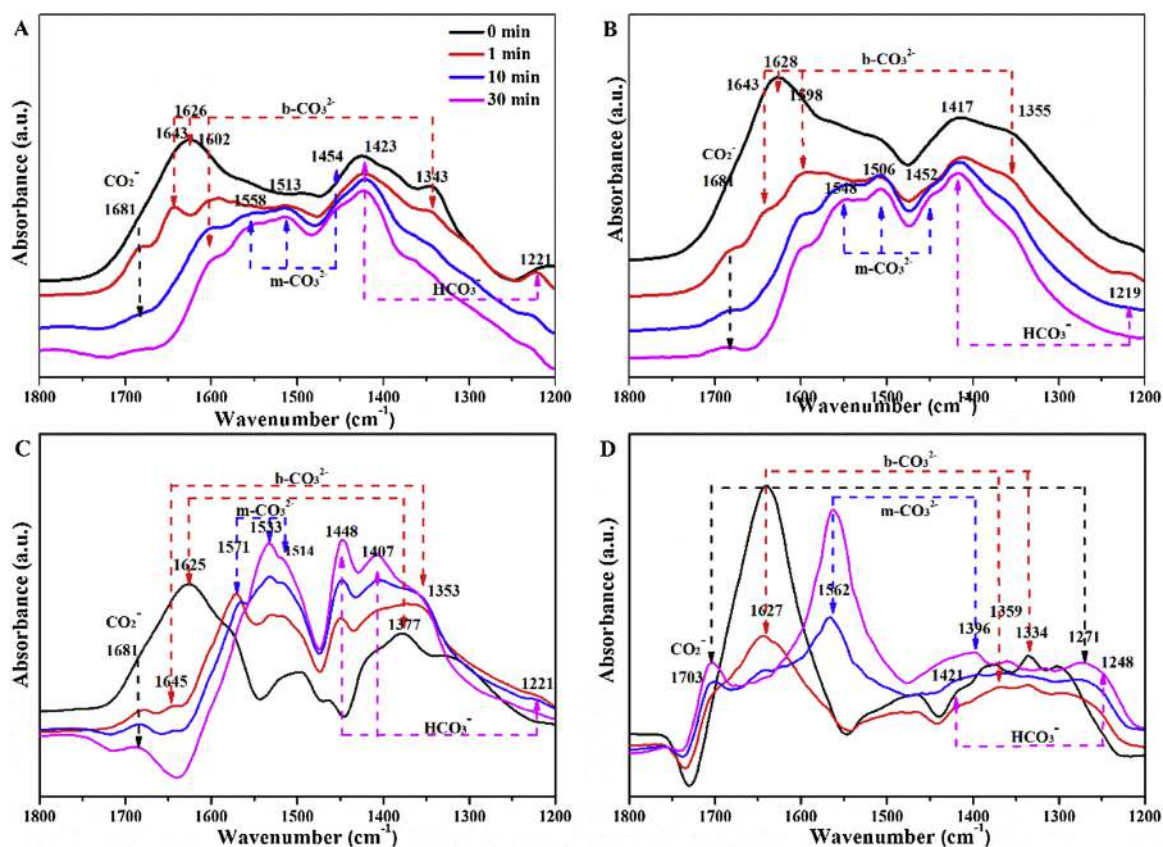


Fig. 8. In situ FTIR spectra of CO_2 under dark and UV irradiation over (A) I-Pt/U-TiO₂-B, (B) P-Pt/U-TiO₂-B, (C) I-Pt/U-TiO₂-U and (D) P-Pt/U-TiO₂-U.

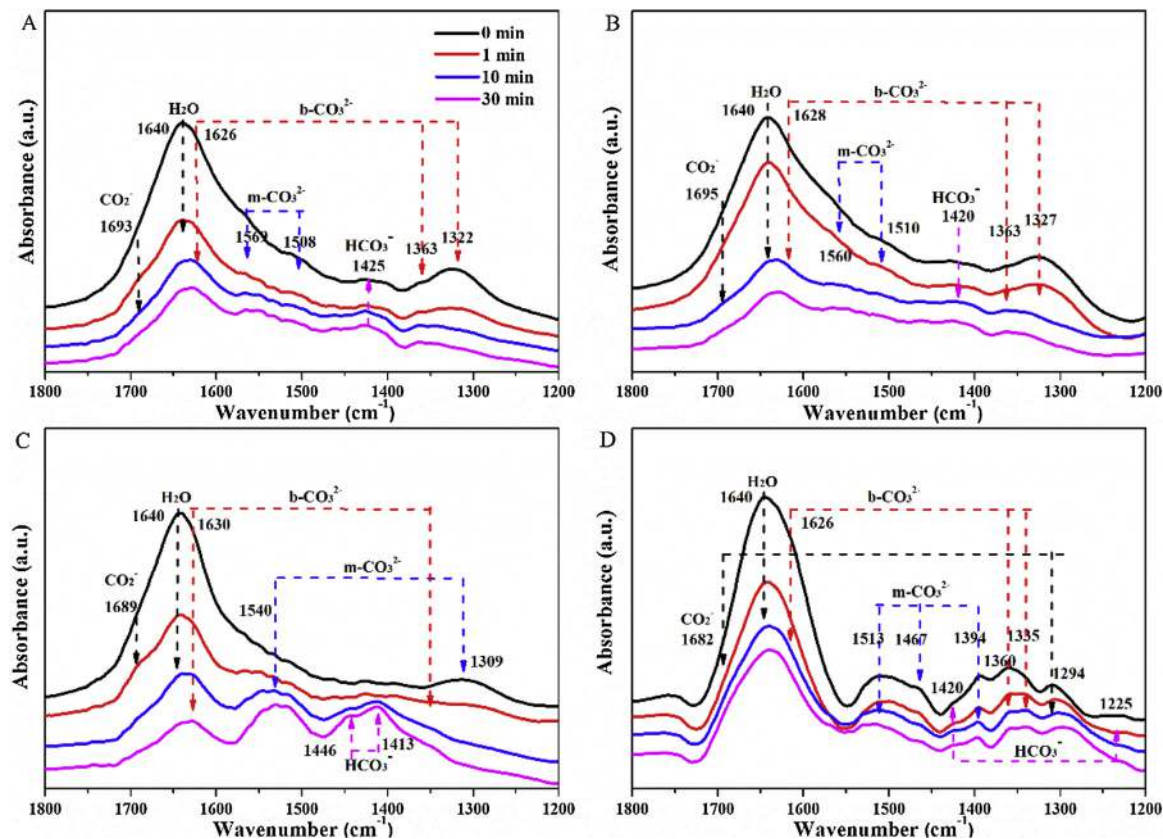


Fig. 9. In situ FTIR spectra of CO_2 with H_2O under dark and UV irradiation over (A) I-Pt/U-TiO₂-B, (B) P-Pt/U-TiO₂-B, (C) I-Pt/U-TiO₂-U and (D) P-Pt/U-TiO₂-U.

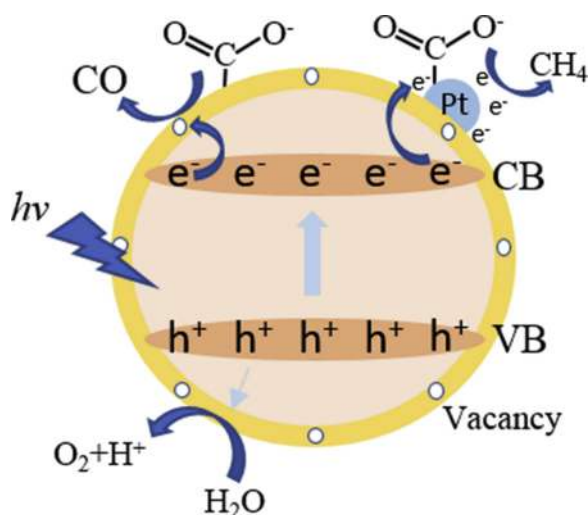


Fig. 10. Possible mechanisms of the ultrathin TiO₂ supported highly dispersed Pt nanoparticles for photoreduction of CO₂ with H₂O vapour.

catalysts with large metal particles. The separation behaviours of charge carriers were further affirmed by the lifetime analysis based on the fluorescence decay shown in Fig. 7D, which were fitted using three exponential functions to obtain the satisfactory residuals ($\chi^2 \approx 1$). The average lifetime (τ_{av}) is calculated based on the equation [64]: $\tau_{av} = \sum B_i \tau_i^2 / \sum B_i \tau_i$, where τ_i is the lifetime, and B_i is the fractional weights. The obtaining average lifetime of these samples decreases in the order of P-Pt/TiO₂-U (143.9 ns) > I-Pt/TiO₂-U (120.2 ns) > P-Pt/TiO₂-B (39.8 ns) > I-Pt/TiO₂-B (32.0 ns) > TiO₂-U (29.2 ns) > TiO₂-B (27.4 ns), in which a much longer lifetime of P-Pt/TiO₂-U offers further evidence of the more effective separation and migration of the pairs of electron and hole [65]. Additionally, due to the electron transfer between Pt nanoparticles and TiO₂, a fast decay in the short-live lifetime scale is observed in the Pt catalysts (Table S3), which corresponds well to XPS result.

Except for the separation of electron-hole pairs, the adsorption and activation of the stable CO₂ molecules are also a vital issue for enhancing CO₂ reduction efficiency [66]. By analysing the results of CO₂ chemisorption (Table 3), it is noted that the amount of chemisorbed CO₂ on ultrathin TiO₂ is higher than bulk TiO₂ and the corresponding supported Pt catalysts, indicating that the ultralarge surface area, abundant defects and the EG-functionalized surface can markedly improve the adsorption of CO₂ on the surfaces of samples. The maximum adsorption capability over P-Pt/TiO₂-U is estimated to be 52.2 $\mu\text{mol g}^{-1}$, higher in comparison of the others, illustrating the Pt catalysts with low-coordinative sites give a further enhancement of adsorption ability.

To verify the role of the sites in Pt catalysts on the adsorption and activation of CO₂, in situ FTIR of CO₂ photoreduction were conducted on defect-poor TiO₂-B supported Pt catalysts and defect-rich TiO₂-U supported Pt catalysts. As shown in the Fig. 8, in the dark, the exposure of CO₂ on the surface of I-Pt/TiO₂-B results in the generation of bidentate carbonate (b-CO_3^{2-}) at 1602–1643 and 1343 cm^{-1} , monodentate carbonate (m-CO_3^{2-}) at 1558 and 1454 cm^{-1} , and bicarbonate (HCO_3^-) at 1423 and 1221 cm^{-1} [67]. Moreover, the CO₂⁻ intermediates at 1681 cm^{-1} bonded with Ti⁴⁺ of supports is also noticed, demonstrating that CO₂ can be trapped at Ti³⁺ sites [68]. Similar CO₂ adsorption species are also observed at the catalysts. However, the band locations of HCO₃⁻ are slightly different. Relative to the impregnated Pt/TiO₂-B and Pt/TiO₂-U as well as photoreduced Pt/TiO₂-B, the normalized peak area of CO₂ intermediates is much larger on P-Pt/TiO₂-U, revealing that CO₂ can be easily adsorbed and activated. Upon subsequent photoillumination, an increase of peak associated with CO₂ intermediates suggests that photoillumination significantly enhances

the ability of activation and dissociation of CO₂ on catalysts under UV light [69].

Furthermore, in situ DRIFTS spectra of CO₂ with H₂O were shown in Fig. 9. The exposure of four Pt catalysts to CO₂ with H₂O vapour in the dark contributes to the dominant features of strongly adsorbed H₂O at 1640 cm^{-1} with relative weak bicarbonate (HCO_3^-) at 1425 cm^{-1} , bidentate carbonates (b-CO_3^{2-}) at 1626 and 1320–1360 cm^{-1} as well as monodentate carbonates (m-CO_3^{2-}) at 1500–1570 cm^{-1} . This depicts that the adsorption of H₂O competes with CO₂ on the surface of catalysts. When the UV irradiation is introduced, no obvious change for the features of CO₂ intermediates and H₂O-related species is seen on the I-Pt/U-TiO₂-B and P-Pt/U-TiO₂-B in comparison with those in the dark, even prolong the time of illumination to 30 min, accounting for the low reaction rate. As expected, significantly different results are observed on the I-Pt/TiO₂-U and P-Pt/TiO₂-U, in which CO₂⁻ species are formed and the intensity of CO₂ intermediates including the carbonates and bicarbonate decreases, indicating the effective activation of CO₂ and H₂O on the ultrathin TiO₂ supported Pt catalysts.

Summarizing the above results, a possible mechanism of ultrathin TiO₂ and highly dispersed Pt nanoparticles for accelerating the efficiency of photoreduction of CO₂ is proposed in Fig. 10. CO₂ is chemisorbed on the Pt catalysts in the manner of bent CO₂⁻, bicarbonate and carbonates, the reactivity of which is considered to be higher than that of the linear adsorbed CO₂. The generated electrons by UV irradiation on ultrathin TiO₂ are easily trapped by the highly dispersed Pt nanoparticles with the terrace owing to the low level of Fermi energy. This electron-rich Pt nanoparticles allow the reduction of CO₂ to produce CH₄. On the other hand, considering the abundant defects (Ti³⁺-V_O) of ultrathin TiO₂ are formed, which are favourable to adsorb the reactants and inhibit the photoinduced charge recombination, the ultrathin TiO₂ may be also considered as the reactive sites to produce CO.

4. Conclusion

Overall, ultrathin TiO₂ nanosheets with thicknesses of the two atomic layers supported the highly stable dispersed Pt catalysts have been synthesized using facile photochemical routes, with I-Pt/TiO₂-U, P-Pt/TiO₂-B and I-Pt/TiO₂-B as control samples, to develop the efficient photocatalysts for the reduction of CO₂ with H₂O. The two-atoms-thick TiO₂ nanosheets were firstly confirmed by HRTEM and AFM. The results of XPS, PAS and ESR further provided the evidence that as the thickness of TiO₂ decreasing defects were created, facilitating the high dispersion of active metal. As expected, the ultrathin TiO₂ supported well-dispersed Pt nanoparticles exhibited excellent photocatalytic activity for the conversion of CO₂ into CH₄ and CO with 666.6 $\mu\text{mol g}^{-1} \text{h}^{-1}$ of total electronic yield in the presence of water vapour. More importantly, after three cycling tests, this sample also showed superior operational stability. This enhanced performance was attributed to the following factors: (1) the ultrathin TiO₂ nanosheets with abundant defects and ultralarge surface area promoted the rate of electron-transfer; (2) the ultrafine Pt nanoparticles facilitated to the separation of photogenerated electrons-holes, and thus improved the light-harvesting capacity and quantum efficiency based on the analysis of XPS, PAS, CO-IR, UV-vis DRS and PL spectroscopy; (3) the synergy of metal and support improved the adsorption ability of CO₂. This work offers deep insights for the design of highly efficient catalysts with coordinatively unsaturated sites for CO₂ photoconversion in the presence of H₂O.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.apcatb.2018.12.028>.

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